Mean Curvature Equations—Interior Gradient Estimate*

Qi Zhou

Lanzhou University

Email:zhouq2020@lzu.edu.cn

First,we omit some fundamental concepts about hypersurface,such as,the first fundamental form,the second fundamental form,principal curvature,mean curvature and Gauss curvature.Refer to [1].

♠ Mean curvature of graphs in $\mathbb{R}^{n+1}$.

THEOREM 1 *Let Ω be a domain in $\mathbb{R}^n$ and Σ be a graph in $\mathbb{R}^{n+1}$ given by $x_{n+1} = u(x)$ for some $u \in C^2(\Omega)$.Then,the mean curvature $H(x)$ of Σ at $(x, u(x))$ corresponding to the upward unit normal vector of Σ is given by*

$$H(x) = \frac{1}{\sqrt{1+|Du(x)|^2}}\left(\Delta u(x) - \frac{u_i(x)u_j(x)}{1+|Du(x)|^2}u_{ij}(x)\right).$$

PROOF We set

$$\mathbf{r}(x) = (x, u(x)).$$

Then,for $i, j = 1, \cdots, n$,

$$\mathbf{r}_i = (\mathbf{e}_i, u_i), \quad \mathbf{r}_{ij} = (0, u_{ij}),$$

where $\mathbf{e}_i$ is the unit vector along the x_i-axis.
Now,we take the upward unit normal vector of Σ that

$$\nu = \frac{1}{\sqrt{1+|Du|^2}}(-Du, 1).$$

Then,

$$g_{ij} = \mathbf{r}_i \cdot \mathbf{r}_j = \delta_{ij} + u_i u_j,$$
$$h_{ij} = \mathbf{r}_{ij} \cdot \nu = \frac{u_{ij}}{\sqrt{1+|Du|^2}}.$$

Straightforward calculation,we have

$$g^{ij} = \delta_{ij} - \frac{u_i u_j}{1+|Du|^2}.$$

We have

$$H = g^{ij} \cdot h_{ij} = \frac{1}{\sqrt{1+|Du|^2}}\left(\Delta u - \frac{u_i u_j}{1+|Du|^2}u_{ij}\right).$$

□

♠ Motivation.

*Nonlinear Elliptic Equations Seminar

Writing the mean curvature equation as

$$a_{ij}(Du)u_{ij} = H(x)\sqrt{1+|Du|^2} \quad \text{in } \Omega,$$

where,for any $p \in \mathbb{R}^n$,

$$a_{ij}(p) = \delta_{ij} - \frac{p_i p_j}{1+|p|^2}.$$

This is the quasilinear equations.
We will calculate the eigenvalues of matrix $A = (a_{ij}(p))_{n\times n}$.Since

$$\begin{aligned}
\det(\lambda I - A) &= \det\left((\lambda-1)I + \begin{pmatrix} \frac{p_1}{1+|p|^2} \\ \vdots \\ \frac{p_n}{1+|p|^2} \end{pmatrix}(p_1,\cdots,p_n)\right) \\
&= \det((\lambda-1)I)\det\left(1 + (p_1,\cdots,p_n)((\lambda-1)I)^{-1}\begin{pmatrix} \frac{p_1}{1+|p|^2} \\ \vdots \\ \frac{p_n}{1+|p|^2} \end{pmatrix}\right) \\
&= (\lambda-1)^n\left(1 + \frac{|p|^2}{(\lambda-1)(1+|p|^2)}\right) \\
&= (\lambda-1)^n + \frac{(\lambda-1)^{n-1}|p|^2}{1+|p|^2},
\end{aligned}$$

set

$$\det(\lambda I - A) = 0,$$

we have

$$\lambda_1 = \cdots = \lambda_{n-1} = 1, \quad \lambda_n = \frac{1}{1+|p|^2}.$$

Therefore,for any $p \in \mathbb{R}^n$ and $\xi \in \mathbb{R}^n$,

$$\frac{1}{1+|p|^2}|\xi| \le a_{ij}(p)\xi_i\xi_j \le |\xi|^2.$$

It is obvious that the mean curvature equation has its ellipticity constants determined by the gradient of solutions.Then,these ellipticity constants are controlled only after the C^1-norms of solutions are derived.

♠ Interior Gradient Estimate.

THEOREM 2 (Interior Gradient Estimate) *Suppose that u is an $L^\infty(B_R)\cap C^3(B_R)$-solution of*

$$\Delta u - \frac{u_i u_j}{1+|Du|^2}u_{ij} = H\sqrt{1+|Du|^2} \quad \text{in } B_R,$$

for some $H \in C^1(B_R)$ with $|H|_{C^1(B_R)} < \infty$.Then,

$$|Du(x_0)| \le \exp\left(C\left(1+\frac{\omega^2}{R^2}\right) + C\left(\omega + \frac{\omega^2}{R}\right)|H|_{L^\infty(B_R)} + C\omega^2|DH|_{L^\infty(B_R)}\right),$$

where $\omega = \sup\limits_{B_R} u - \inf\limits_{B_R} u$ and C is a positive constant depending only on n.

PROOF Since

$$\Delta u - \frac{u_i u_j}{1+|Du|^2} u_{ij} = H\sqrt{1+|Du|^2} \Longrightarrow \left(\delta_{ij} - \frac{u_i u_j}{1+|Du|^2}\right) u_{ij} = H\sqrt{1+|Du|^2},$$

then set,for any $x \in \Omega$ and any $p \in \mathbb{R}^n$,

$$a_{ij}(p) = \delta_{ij} - \frac{p_i p_j}{1+|p|^2}$$

and

$$f(x,p) = H(x)\sqrt{1+|p|^2}.$$

Then,we write the mean curvature equations as

$$a_{ij}(Du)u_{ij} = f(x, Du) \quad \text{in } B_R.$$

We assume $u \geq 0$.
Let γ and ψ be two positive functions defined in $[0,\infty)$,with positive derivatives.These functions will be determined.Let η be a nonnegative function in B_R with $\eta = 0$ on ∂B_R.
Set

$$v = \eta\gamma(u)\psi(|Du|^2) \quad \text{in } B_R.$$

We assume v attains its maximum at $x_0 \in B_R$.Then,

$$\eta\gamma(u)\psi(|Du|^2) \leq \eta(x_0)\gamma(u(x_0))\psi(|Du(x_0)|^2) \quad \text{in } B_R.$$

In the following,we assume $\eta(x_0) \neq 0$.Then,

$$(\log v)_i(x_0) = 0, \quad ((\log v)_{ij}(x_0)) \leq 0.$$

Now,we evaluate $a_{ij}(\log v)_{ij}$ at x_0.
Straightforward to calculate,we have

$$(\log v)_i = \frac{2\psi'}{\psi} u_k u_{ki} + \frac{\gamma'}{\gamma} u_i + \frac{\eta_i}{\eta}$$

and

$$\begin{aligned}(\log v)_{ij} =& \frac{2\psi'}{\psi}(u_k u_{kij} + u_{ki}u_{kj}) + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right) u_k u_l u_{ki} u_{lj} \\ &+ \frac{\gamma'}{\gamma} u_{ij} + \left(\frac{\gamma''}{\gamma} - \frac{(\gamma')^2}{\gamma^2}\right) u_i u_j + \left(\frac{\eta_{ij}}{\eta} - \frac{\eta_i \eta_j}{\eta^2}\right).\end{aligned}$$

Hence,

$$\begin{aligned}a_{ij}(\log v)_{ij} =& \frac{2\psi'}{\psi}(u_k a_{ij} u_{kij} + a_{ij} u_{ki} u_{kj}) + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right) a_{ij} u_k u_l u_{ki} u_{lj} \\ &+ \frac{\gamma'}{\gamma} a_{ij} u_{ij} + \left(\frac{\gamma''}{\gamma} - \frac{(\gamma')^2}{\gamma^2}\right) a_{ij} u_i u_j + a_{ij}\left(\frac{\eta_{ij}}{\eta} - \frac{\eta_i \eta_j}{\eta^2}\right).\end{aligned}$$

• **Eliminating the third derivatives of u in the expression above.**

We differentiate $a_{ij}(Du)u_{ij} = f(x, Du)$.

$$
\begin{aligned}
&(a_{ij}(Du)u_{ij})_k = D_k f \\
\Longrightarrow &(a_{ij}(Du))_k u_{ij} + a_{ij}(Du)u_{ijk} = D_k f \\
\Longrightarrow &\left(\delta_{ij} - \frac{u_i u_j}{1+|Du|^2}\right)_k u_{ij} + a_{ij}(Du)u_{ijk} = D_k f \\
\Longrightarrow &\left(-\frac{u_{ki}u_j + u_{kj}u_i}{1+|Du|^2} + \frac{2u_i u_j u_l u_{kl}}{(1+|Du|^2)^2}\right) u_{ij} + a_{ij}(Du)u_{ijk} = D_k f \\
\Longrightarrow &\left(-\frac{\delta_{li}p_j + \delta_{lj}p_i}{1+|p|^2} + \frac{2p_i p_j p_l}{(1+|p|^2)^2}\right) u_{ij}u_{kl} + a_{ij}(Du)u_{ijk} = D_k f \\
\Longrightarrow &a_{ij}u_{kij} + a_{ij,p_l}u_{kl}u_{ij} = D_k f,
\end{aligned}
$$

where

$$a_{ij,p_l} = -\frac{\delta_{li}p_j + \delta_{lj}p_i}{1+|p|^2} + \frac{2p_i p_j p_l}{(1+|p|^2)^2}.$$

A simple substitution yields

$$a_{ij}(\log v)_{ij} = I + II + \left(\frac{\gamma''}{\gamma} - \frac{(\gamma')^2}{\gamma^2}\right) a_{ij}u_i u_j + a_{ij}\left(\frac{\eta_{ij}}{\eta} - \frac{\eta_i \eta_j}{\eta^2}\right),$$

where

$$I = \frac{2\psi'}{\psi}(-u_k a_{ij,p_l}u_{kl}u_{ij} + a_{ij}u_{ki}u_{kj}) + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right) a_{ij}u_k u_l u_{ki}u_{lj}$$

and

$$II = \frac{2\psi'}{\psi}u_k D_k f + \frac{\gamma'}{\gamma}f.$$

In the following,all calculations are made at x_0.

• **Eliminating the second derivatives of u in I and II.**

To simplify our calculation,we first choose an appropriate coordinate such that

$$Du(x_0) = (u_1(x_0), 0, \cdots, 0),$$

with $u_1(x_0) > 0$.In particular,for $i = 2, \cdots, n$,

$$u_i(x_0) = 0.$$

Hence,

$$I = \frac{2\psi'}{\psi}(-u_1 a_{ij,p_l}u_{1l}u_{ij} + a_{ij}u_{ki}u_{kj}) + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right) a_{ij}u_1^2 u_{1i}u_{1j}.$$

Moreover

$$
\begin{aligned}
&a_{11} = 1 - \frac{u_1^2}{1+u_1^2} = \frac{1}{1+u_1^2}, \quad a_{ii} = 1(i \neq 1), \quad a_{ij} = 0(i \neq j), \\
&a_{11,p_1} = -\frac{u_1 + u_1}{1+u_1^2} + \frac{2u_1u_1u_1}{(1+u_1^2)^2} = -\frac{2u_1}{1+u_1^2} + \frac{2u_1^3}{(1+u_1^2)^2} = -\frac{2u_1}{(1+u_1^2)^2}, \\
&a_{11,p_l} = 0(l \neq 1), \quad a_{1j,p_j} = -\frac{u_1}{1+u_1^2}(j \neq 1), \quad a_{1j,p_l} = 0(j \neq 1, l \neq j) \\
&a_{ij,p_l} = 0(i \neq 1, j \neq 1).
\end{aligned}
$$

In fact,we can write

$$a_{11,p_1} = -\frac{2u_1}{1+u_1^2}a_{11}$$

and,for any $i = 2, \cdots, n$,

$$a_{1i,p_i} = -\frac{u_1}{1+u_1^2}a_{ii}.$$

Then,

$$\begin{aligned}
-u_1 a_{ij,p_l} u_{1l} u_{ij} &= -u_1 a_{11,p_1} u_{11}^2 - 2\sum_{j=2}^{n} u_1 a_{1j,p_j} u_{1j}^2 \\
&= \frac{2u_1^2}{1+u_1^2} a_{11} u_{11}^2 + \sum_{j=2}^{n} \frac{2u_1^2}{1+u_1^2} a_{jj} u_{1j}^2 \\
&= \frac{2u_1^2}{1+u_1^2} \sum_{i=1}^{n} a_{ii} u_{1i}^2.
\end{aligned}$$

Hence,

$$\begin{aligned}
I =& \frac{2\psi'}{\psi}\left(\frac{2u_1^2}{1+u_1^2}\sum_{i=1}^{n} a_{ii}u_{1i}^2 + \sum_{i,k=1}^{n} a_{ii}u_{ki}^2\right) + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right)\sum_{i=1}^{n} a_{ii}u_1^2 u_{1i}^2 \\
=& \frac{2\psi'}{\psi}\left(\frac{2u_1^2}{1+u_1^2}\left(\frac{u_{11}^2}{1+u_1^2} + \sum_{i=2}^{n} u_{1i}^2\right) + \frac{1}{1+u_1^2}u_{11}^2 + \frac{1}{1+u_1^2}\sum_{k=2}^{n} u_{k1}^2 + \sum_{i,k=2}^{n} u_{ki}^2 + \sum_{i=2}^{n} u_{1i}^2\right) \\
&+ 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right)\left(\frac{1}{1+u_1^2}\cdot u_1^2 u_{11}^2 + \sum_{i=2}^{n} u_1^2 u_{1i}^2\right) \\
=& \frac{2\psi'}{\psi}\left(\left(\frac{2u_1^2}{1+u_1^2} + 1\right)\frac{u_{11}^2}{1+u_1^2} + \frac{2u_1^2}{1+u_1^2}\sum_{i=2}^{n} u_{1i}^2 + \frac{1}{1+u_1^2}\sum_{k=2}^{n} u_{k1}^2 + \sum_{i,k=2}^{n} u_{ki}^2 + \sum_{i=2}^{n} u_{1i}^2\right) \\
&+ 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right)\left(\frac{u_{11}^2}{1+u_1^2}\cdot u_1^2 + \sum_{i=2}^{n} u_1^2 u_{1i}^2\right) \\
=& \left(\frac{2\psi'}{\psi}\cdot\frac{1+3u_1^2}{1+u_1^2} + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right)u_1^2\right)\frac{u_{11}^2}{1+u_1^2} + \left(\frac{2\psi'}{\psi}\cdot\frac{2+3u_1^2}{1+u_1^2} + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right)u_1^2\right)\sum_{i=2}^{n} u_{1i}^2 \\
&+ \frac{2\psi'}{\psi}\sum_{i,k=2}^{n} u_{ik}^2.
\end{aligned}$$

Since

$$\begin{aligned}
D_1 f &= \left(H\sqrt{1+|Du|^2}\right)_1 \\
&= H_1\sqrt{1+u_1^2} + H\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{1+u_1^2}}(1+|Du|^2)_1 \\
&= H_1\sqrt{1+u_1^2} + \frac{1}{2}\cdot\frac{H}{\sqrt{1+u_1^2}}\cdot 2u_1 u_{11} \\
&= H_1\sqrt{1+u_1^2} + \frac{H}{\sqrt{1+u_1^2}}u_1 u_{11}.
\end{aligned}$$

Then

$$
\begin{aligned}
II &= \frac{2\psi'}{\psi}u_1\left(H_1\sqrt{1+u_1^2}+\frac{H}{\sqrt{1+u_1^2}}u_1u_{11}\right)+\frac{\gamma'}{\gamma}H\sqrt{1+u_1^2}\\
&= \frac{2\psi'}{\psi}u_1\left(H_1\sqrt{1+u_1^2}+\frac{H}{\sqrt{1+u_1^2}}u_1u_{11}\right)+\frac{\gamma'}{\gamma}\frac{H}{\sqrt{1+u_1^2}}\cdot(1+u_1^2)\\
&= \frac{H}{\sqrt{1+u_1^2}}\left(\frac{\gamma'}{\gamma}(1+u_1^2)+\frac{2\psi'}{\psi}u_1^2u_{11}\right)+\frac{2\psi'}{\psi}u_1H_1\sqrt{1+u_1^2}.
\end{aligned}
$$

Since $(\log v)_i=\dfrac{2\psi'}{\psi}u_ku_{ki}+\dfrac{\gamma'}{\gamma}u_i+\dfrac{\eta_i}{\eta}$ and $v_i(x_0)=0$,then

$$
\frac{2\psi'}{\psi}u_1u_{1i}=-\left(\frac{\gamma'}{\gamma}u_i+\frac{\eta_i}{\eta}\right).
$$

Hence

$$
\begin{cases}
u_{ii}=-\dfrac{\psi}{2\psi'}\left(\dfrac{\gamma'}{\gamma}+\dfrac{\eta_1}{u_1\eta}\right) & i=1\\
u_{1i}=-\dfrac{\psi}{2\psi'}\cdot\dfrac{\eta_i}{u_1\eta} & i=2,\cdots,n.
\end{cases}
$$

Therefore,

$$
\begin{aligned}
I=&\left(\frac{2\psi'}{\psi}\cdot\frac{1+3u_1^2}{1+u_1^2}+4\left(\frac{\psi''}{\psi}-\frac{(\psi')^2}{\psi^2}\right)u_1^2\right)\frac{\psi^2}{4(1+u_1^2)(\psi')^2}\left(\frac{\gamma'}{\gamma}+\frac{\eta_1}{u_1\eta}\right)^2\\
&+\left(\frac{2\psi'}{\psi}\cdot\frac{2+3u_1^2}{1+u_1^2}+4\left(\frac{\psi''}{\psi}-\frac{(\psi')^2}{\psi^2}\right)u_1^2\right)\frac{\psi^2}{4u_1^2(\psi')^2}\sum_{i=2}^n\frac{\eta_i^2}{\eta^2}+\frac{2\psi'}{\psi}\sum_{i,k=2}^n u_{ik}^2
\end{aligned}
$$

and

$$
II=\frac{H}{\sqrt{1+u_1^2}}\left(\frac{\gamma'}{\gamma}-\frac{\eta_1}{\eta}u_1\right)+\frac{2\psi'}{\psi}u_1H_1\sqrt{1+u_1^2}.
$$

We now choose γ and ψ.Set $M=\sup\limits_{B_R}u$ and

$$
\gamma(t)=1+\frac{t}{M},\quad \psi(t)=\log t.
$$

Then,

$$
\gamma'(t)=\frac{1}{M},\quad \gamma''(t)=0,
$$

and

$$
\psi'(t)=\frac{1}{t},\quad \psi''(t)=-\frac{1}{t^2}.
$$

Hence,$1\le\gamma(u)\le 2$ since $0\le u\le M$,and

$$
\frac{\psi'}{\psi}(|Du|^2)=\frac{1}{2u_1^2\log u_1},\quad \frac{\psi''}{\psi}(|Du|^2)=-\frac{1}{2u_1^4\log u_1}.
$$

• For I.

A straightforward calculation yields

$$
\frac{2\psi'}{\psi}\cdot\frac{1+3u_1^2}{1+u_1^2}+4\left(\frac{\psi''}{\psi}-\frac{(\psi')^2}{\psi^2}\right)u_1^2=\frac{1}{u_1^2\log u_1}\left(1-\frac{2}{1+u_1^2}-\frac{1}{\log u_1}\right).
$$

Hence,if $u_1 \geq e^3$,then

$$\frac{2\psi'}{\psi} \cdot \frac{1+3u_1^2}{1+u_1^2} + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right) u_1^2 \geq \frac{1}{2u_1^2 \log u_1} > 0.$$

Therefore

$$\left(\frac{2\psi'}{\psi} \cdot \frac{2+3u_1^2}{1+u_1^2} + 4\left(\frac{\psi''}{\psi} - \frac{(\psi')^2}{\psi^2}\right) u_1^2\right) \frac{\psi^2}{4u_1^2(\psi')^2} \sum_{i=2}^{n} \frac{\eta_i^2}{\eta^2} + \frac{2\psi'}{\psi} \sum_{i,k=2}^{n} u_{ik}^2 \geq 0.$$

Using $1 + u_1^2 \leq 2u_1^2$,we obtain

$$I \geq \frac{1}{2u_1^2 \log u_1} \cdot \frac{\psi^2}{8u_1^2(\psi')^2} \left(\frac{\gamma'}{\gamma} + \frac{\eta_1}{u_1 \eta}\right)^2 = \frac{1}{4} \log u_1 \left(\frac{\gamma'}{\gamma} + \frac{\eta_1}{u_1 \eta}\right)^2.$$

• For II.

A straightforward calculation yields

$$\begin{aligned} II &= \frac{H}{\sqrt{1+u_1^2}} \left(\frac{\gamma'}{\gamma} - \frac{\eta_1}{\eta} u_1\right) + \frac{2\psi'}{\psi} u_1 H_1 \sqrt{1+u_1^2} \\ &= \frac{H}{\sqrt{1+u_1^2}} \left(\frac{1}{M+u_1} - \frac{\eta_1}{\eta} u_1\right) + H_1 \frac{\sqrt{1+u_1^2}}{u_1 \log u_1}. \end{aligned}$$

Hence,if $u_1 > e^2$,then

$$II \geq -|H| \left(\frac{1}{M} + \frac{|\eta_1|}{\eta}\right) - |H_1|.$$

We also have

$$\begin{aligned} \left|\left(\frac{\gamma''}{\gamma} - \frac{(\gamma')^2}{\gamma^2}\right) a_{ij} u_i u_j\right| &= \left|\left(\left(1 + \frac{u_1^2}{M}\right) \cdot 0 - \frac{(\gamma')^2}{\gamma^2}\right) a_{11} u_1^2\right| \\ &= \frac{(\gamma')^2}{\gamma^2} a_{11} u_1^2 = \frac{u_1^2}{1+u_1^2} \left(\frac{1}{M} \cdot \frac{1}{1+u_1/M}\right)^2 \\ &\leq \frac{1}{M^2}. \end{aligned}$$

Since $a_{ij}(\log v)_{ij}(x_0) \leq 0$,we obtain,at x_0,

$$\frac{1}{4} \log u_1 \left(\frac{\gamma'}{\gamma} + \frac{\eta_1}{u_1 \eta}\right)^2 \leq a_{ij} \left(\frac{\eta_i \eta_j}{\eta^2} - \frac{\eta_{ij}}{\eta}\right) + \frac{1}{M^2} + |H| \left(\frac{1}{M} + \frac{|\eta_1|}{\eta}\right) + |DH|.$$

We choose η such that

$$|D^2 \eta| + \frac{|D\eta|^2}{\eta} \leq \frac{c}{R^2} \quad \text{in } B_R,$$

for some positive constant c.For example,we can take

$$\eta(x) = \left(1 - \frac{|x|^2}{R^2}\right)^2.$$

Then,

$$\log u_1 \left(\frac{\gamma'}{\gamma} + \frac{\eta_1}{u_1 \eta}\right)^2 \leq C \left(\frac{1}{R^2 \eta} + \frac{1}{M^2} + \left(\frac{1}{M} + \frac{1}{R\sqrt{\eta}}\right) |H| + |DH|\right),$$

where C is a positive constant depending only on n.

We set

$$\widetilde{M} = \frac{M^2}{R^2} + 1 + M\left(1 + \frac{M}{R}\right) |H|_{L^\infty} + M^2 |DH|_{L^\infty}.$$

Since $\eta \in (0,1]$ in B_R,then,if $u_1 > e^3$,

$$\log u_1 \left(\frac{\gamma'}{\gamma} + \frac{\eta_1}{u_1\eta}\right)^2 \leq \frac{C\widetilde{M}}{M^2\eta}.$$

- $$\left|\frac{\eta_1}{u_1\eta}\right| \leq \frac{\gamma'}{2\gamma}.$$

We have

$$\begin{aligned}
&\log u_1 \left(\frac{\gamma'}{\gamma} + \frac{\eta_1}{u_1\eta}\right)^2 \leq \frac{C\widetilde{M}}{M^2\eta} \\
\Longrightarrow &\log u_1 \left(\frac{\gamma'}{\gamma} - \frac{\gamma'}{2\gamma}\right)^2 \leq \frac{C\widetilde{M}}{M^2\eta} \\
\Longrightarrow &\frac{(\gamma')^2}{4\gamma^2} \log u_1 \leq \frac{C\widetilde{M}}{M^2\eta}.
\end{aligned}$$

Since $1 \leq \gamma(u) \leq 2$ in B_R and $(\gamma'(u))^2 = \dfrac{1}{M^2}$,we get

$$\eta\gamma \log u_1 \leq C\widetilde{M}.$$

- $$\frac{\gamma'}{2\gamma} \leq \left|\frac{\eta_1}{u_1\eta}\right|.$$

Since $\eta(x) = \left(1 - \dfrac{|x|^2}{R^2}\right)^2$,then

$$\eta_1 = 2\left(1 - \frac{|x|^2}{R^2}\right)\left(-\frac{1}{R^2} \cdot 2x_1\right) \Longrightarrow |\eta_1| = \left(1 - \frac{|x|^2}{R^2}\right) \cdot \frac{|x_1|}{R^2} \leq \frac{1}{R}.$$

Since $1 \leq \gamma(u) \leq 2$ in B_R and $(\gamma'(u))^2 = \dfrac{1}{M^2}$,we get

$$\frac{\gamma'}{2\gamma} \leq \left|\frac{\eta_1}{u_1\eta}\right| \leq \frac{1}{u_1\eta} \cdot \frac{1}{R} \Longrightarrow \eta u_1 \gamma \leq C\frac{M}{R}.$$

Hence,if $u_1 > e$,

$$\eta\gamma \log u_1 \leq C\frac{M}{R}.$$

In conclusion,we are,at x_0,

$$\eta\gamma \log u_1 \leq C\left(1 + \frac{M^2}{R^2} + \frac{M}{R} + M\left(1 + \frac{M}{R}\right)|H|_{L^\infty} + M^2|DH|_{L^\infty}\right).$$

Since $u_1(x_0) = |Du(x_0)|$ and $u \geq 0$,then

$$\begin{aligned}
&|Du(x_0)| \leq \exp\left(C\left(1 + \frac{M^2}{R^2} + \frac{M}{R}\right) + C\left(M + \frac{M^2}{R}\right)|H|_{L^\infty} + CM^2|DH|_{L^\infty}\right) \\
\Longrightarrow &|Du| \leq \exp\left(C\left(1 + \frac{\omega^2}{R^2}\right) + C\left(\omega + \frac{\omega^2}{R}\right)|H|_{L^\infty(B_R)} + C\omega^2|DH|_{L^\infty(B_R)}\right),
\end{aligned}$$

where $\omega = \sup\limits_{B_R} u - \inf\limits_{B_R} u$. □

References

[1] Qing Han, Nonlinear Elliptic Equations of the Second Order, American Mathematical Society, GSM171, 2016.